



250 YEARS OF RISKS IN FIFTEEN MINUTES

Every breakthrough solved one risk and created the next.

What 250 years predicts about AI on the plant floor.

GRAYSON ESTES
Sept 2026

WHO AM I?

[Grayson Estes]



```
sh-3.2#  
sh-3.2#  
sh-3.2#  
sh-3.2# Whoami>?
```

>>> **Grayson Estes**

>> ✓ 15+ years in InfoSec

>> 8+ years on OT / ICS industrial efforts

✓ Automation Builder → InfoSec Researcher → Product Leader
(Heavy AI and ML background)

✓ Worked across Energy, Manufacturing, Robotics, Utilities, Mil/Gov & critical infrastructure

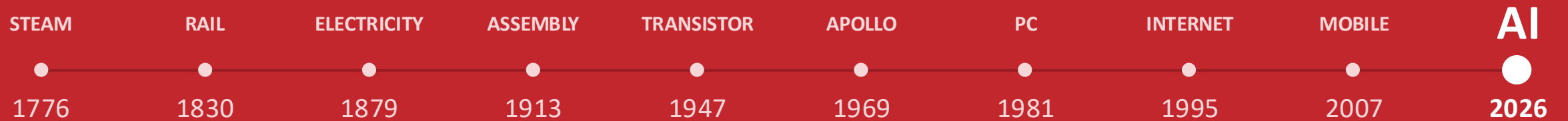
✓ Built and deployed OT & AI cybersecurity technology in real environments

✓ Security researcher — Automotive, UAS, medical devices IoT & more

sh-3.2# why_am_i_here?

>>> I've spent my career where cyber meets the physical world.

250 years. One pattern.



EVERY BREAKTHROUGH WROTE A NEW FAILURE MODE

Risks created in ways unexpected



01

PART ONE

THE ARC OF WORK & WORKFLOW REDESIGN

Value usually never arrived when a new engine was bolted onto an old floor. Value typically arrives when the entire architecture of work is rebuilt around it.

THE HOOK

1840s.
Britain solves rail's
biggest problem.

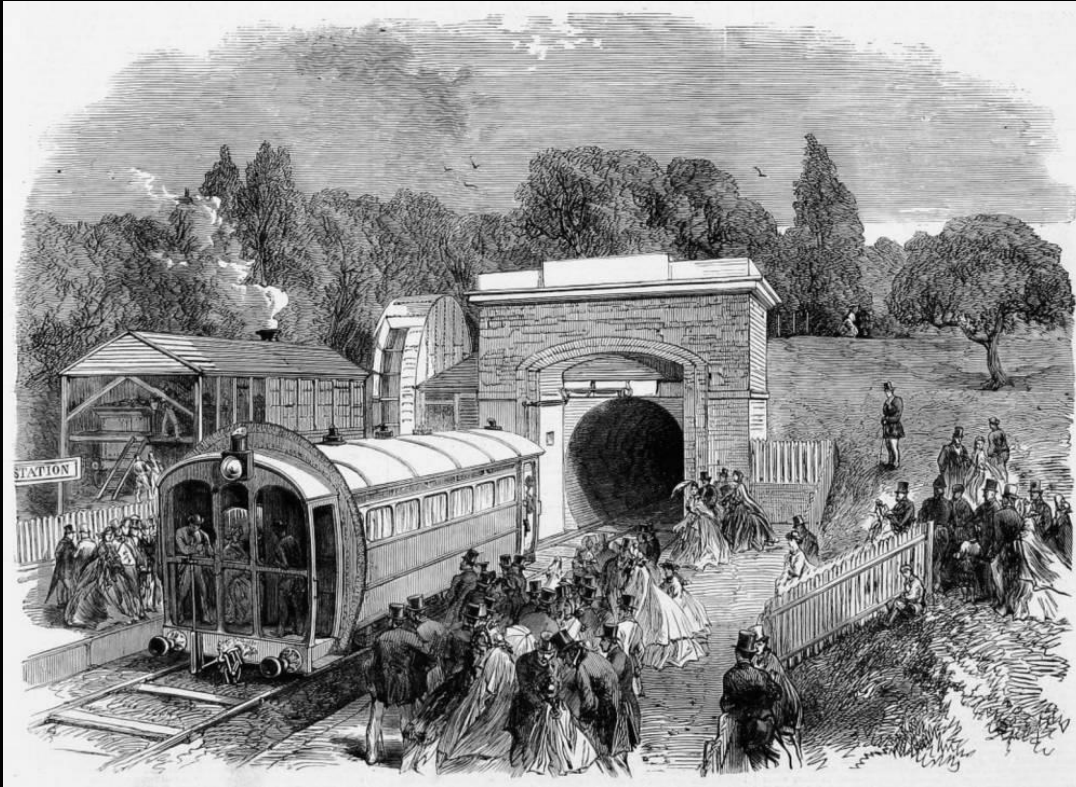


IMAGE SLOT — Brunel's South Devon atmospheric railway: cutaway of the vacuum pipe between the rails, or the surviving Starcross pumping house.

1840s

ATMOSPHERIC RAILWAY

No engine.

No boiler.

No coal.

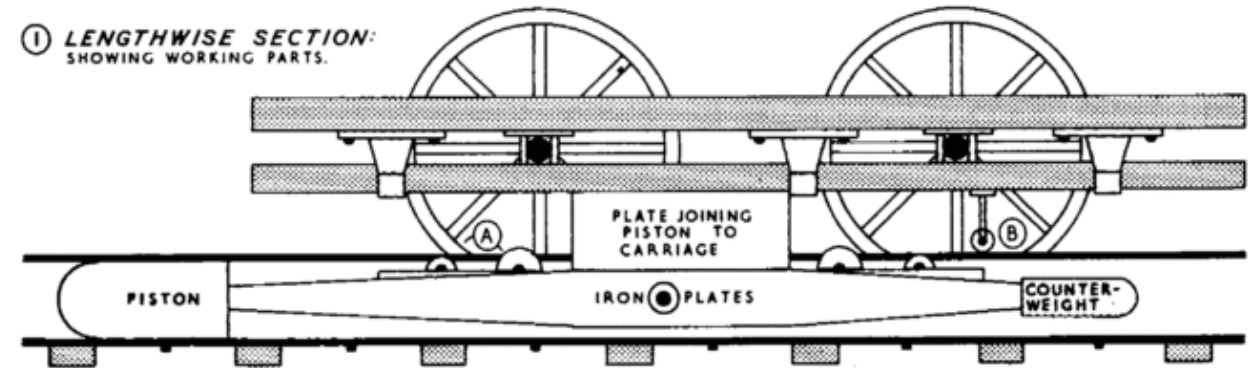
**Just a vacuum,
pulling the train
down a pipe.**

It worked.

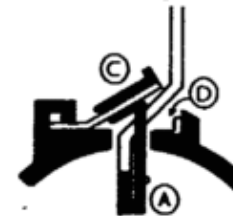
Quieter, cleaner and faster
than anything on rails.

THE ATMOSPHERIC RAILWAY

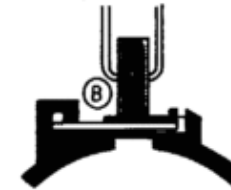
① LENGTHWISE SECTION:
SHOWING WORKING PARTS.



② VALVE SECTIONS:
OPENING



(LARGER SCALE)
CLOSING



③ SECTION OF TUBE AND TRACK:



- A - ROLLER OPENING VALVE
- B - ROLLER CLOSING VALVE
- C - LEATHER VALVE REINFORCED WITH IRON PLATES
- D - GREASE COMPOSITION SEAL

THE RAT-EATEN SEAL: WHY GREAT IDEAS DIE

BRUNEL'S SOUTH DEVON VACUUM RAILWAY

The Real Failure Mode: The physics worked perfectly. But the longitudinal leather flap holding the vacuum degraded under weather, iron corrosion, and rats. The system lacked maintenance context and environmental durability.

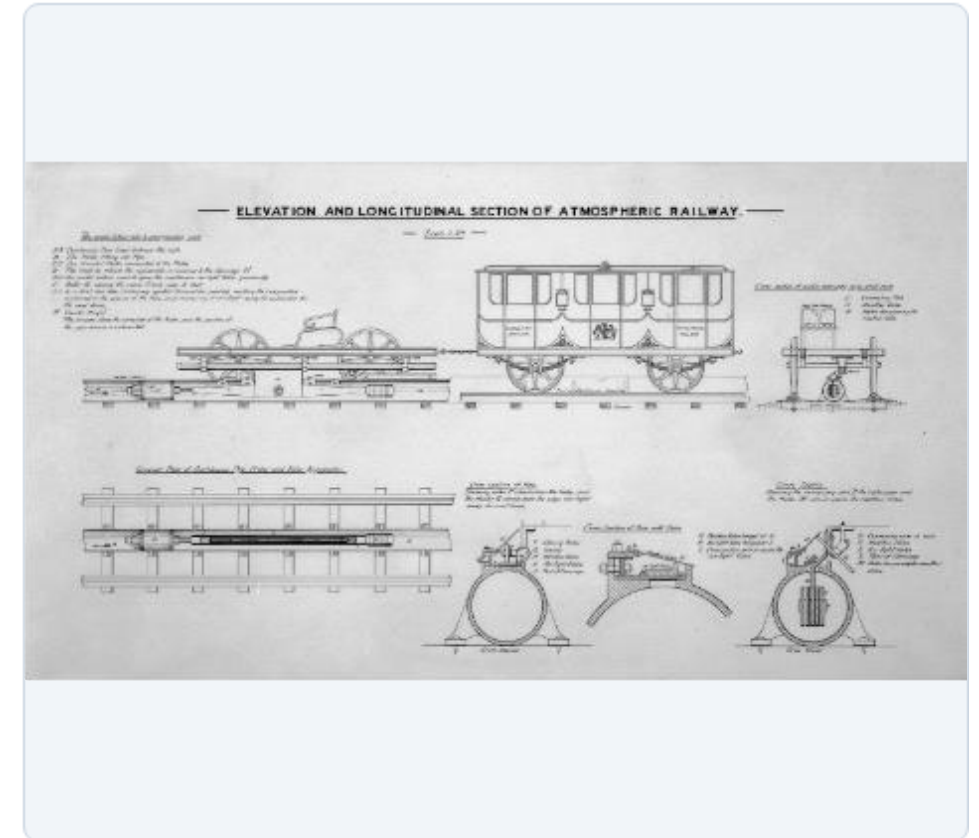




IMAGE SLOT — the longitudinal leather valve on the vacuum pipe. A section survives at Didcot Railway Centre; Google Patents US44376A has period valve drawings.

Root cause

THE SEAL

**Not the physics.
A leather flap
sealed the pipe.**

And it rotted.

02

PART TWO

EVERY REVOLUTION REWRITES RISKs

We do not eliminate danger when technology advances.

We often trade mechanical hazards we can see and touch for systemic vulnerabilities we cannot even define or understand.

THE 40-YEAR GAP: BOLTED-ON VS RE-ARCHITECTED

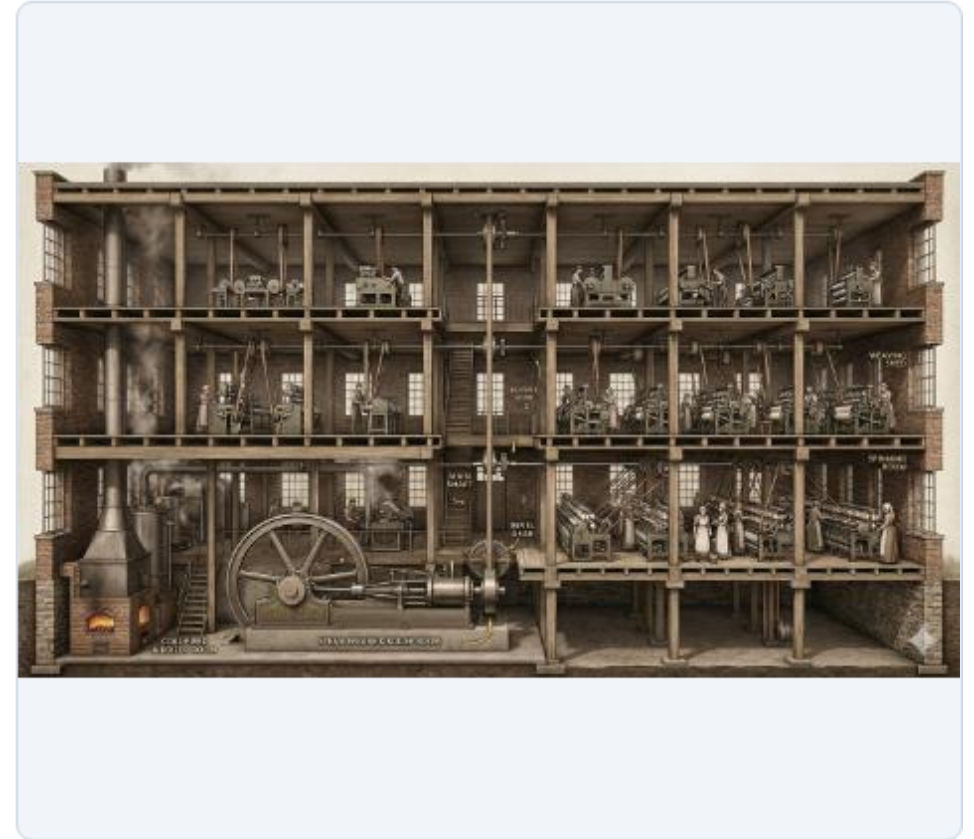
PHASE 1: BOLTED-ON (1890S)

Owners pulled out steam boilers and bolted giant electric motors to the exact same overhead line-shafts. Output rose less than 2%. Electricity was branded overhyped.

PHASE 2: UNIT DRIVE REDESIGN (1920S)

They ripped down the line-shafts, built single-story flat plants, put fractional motors on each tool, and arranged machines in sequential production flow. Output exploded 300%.

Takeaway for AI: Bolting an LLM or agent onto a 1980s SCADA hierarchy gains nothing until the operational floor plan is redesigned.



THE INDUSTRIAL RISK LEDGER: WHAT WE TRADED



What We Solved ELIMINATED BOTTLENECKS

- ✓ **Steam Era:** Eliminated water/muscle dependency and geographic constraints on factories.
- ✓ **Assembly Era:** Standardized skilled labor, lowered cycle times from 12 hours to 93 minutes.
- ✓ **Digital/PLC Era:** Eradicated physical relay wiring and manual mechanical adjustments.
- ✓ **AI & Agents:** Triaging massive telemetry volume and automating microsecond control corrections.



What We Introduced NEW FAILURE MODES

- ✗ **Boiler Explosions:** Placed pressurized bombs directly inside work centers next to humans.
- ✗ **Interdependent Jamming:** Single-station failure collapses the entire line simultaneously.
- ✗ **Cyber-Physical Attack:** Stuxnet (2010) and Triton (2017) weaponized deterministic logic code.
- ✗ **Autonomous Opacity:** Non-deterministic drift, poisoned data, and machine-speed unverified actions.

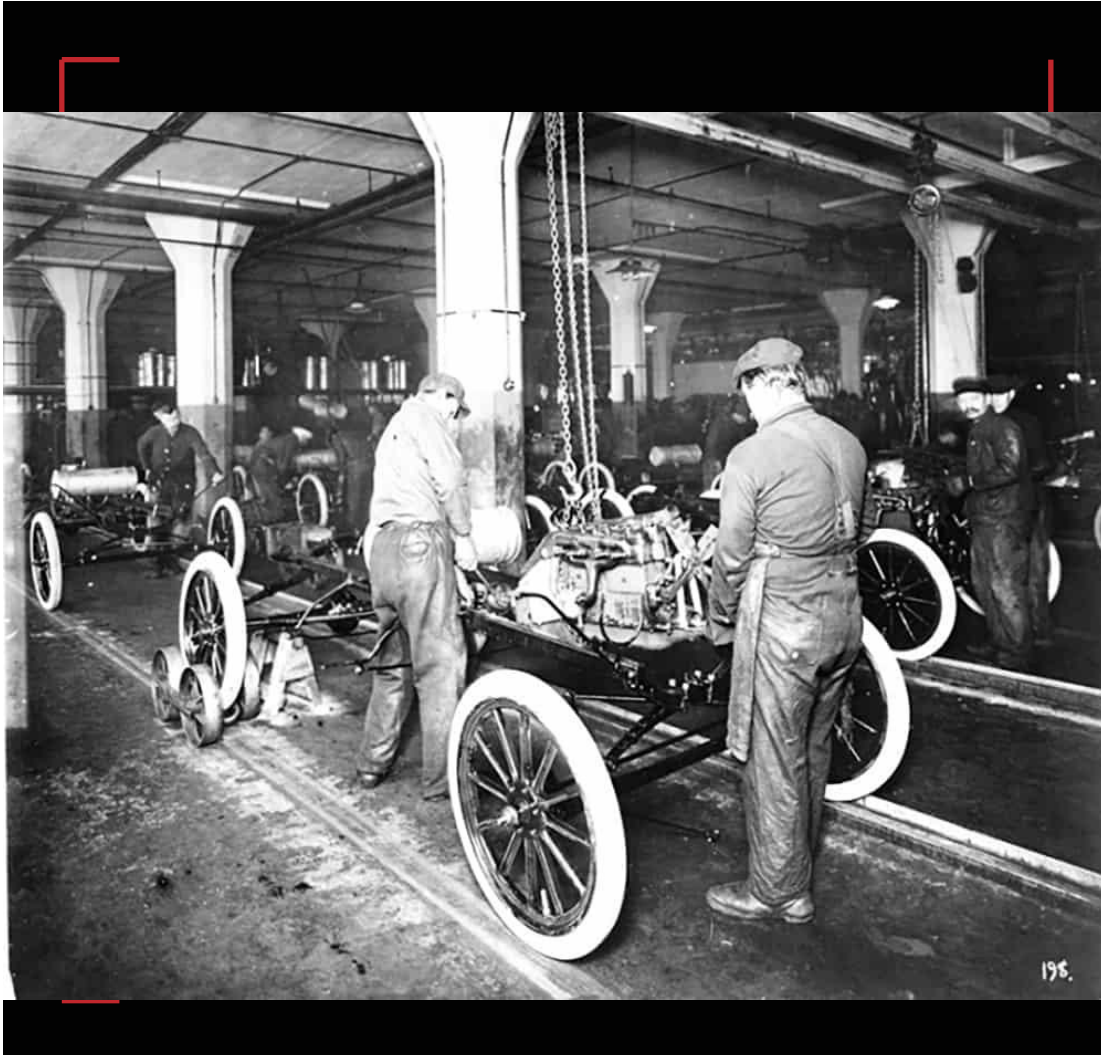


IMAGE SLOT — Ford Highland Park moving chassis line, 1913.

1913

ASSEMBLY LINE

**Twelve hours
to ninety minutes.**

**Nothing new
was invented.**

1913 • The moving line

SOLVED

- The cost of a car (Cheaper to make)
- Skill scarcity per station (Cross Training)
- Throughput ceilings (Shift Work)



INTRODUCED

- Total interdependence — one jam anywhere stops everything everywhere
- A machine-set pace, and its human cost



One of the first times efficiency and fragility became the same design decision.

250 YEARS OF INDUSTRIAL VULNERABILITY

1830s

BOILER EXPLOSIONS

Safety codes emerge only after hundreds of steam engine catastrophic ruptures in public spaces.

1880s

SHOCK & FIRE

Unregulated early electrical wiring brings unseen arc flash and building fires into textile mills.

1913

SINGLE-POINT JAM

The Highland Park moving line trades operator autonomy for complete, rigid plant-wide interdependence.

1988

MORRIS WORM

The first networked buffer-overflow attack shows connected machines can be crashed from afar.

2010

STUXNET

Software crosses into physical physics: altering centrifuge speeds while feeding fake telemetry to operators.

2026

AGENT AUTONOMY

Autonomous AI writes back to control loops with zero reaction time left for human intervention.

HOW RISK BECAME INVISIBLE: MECHANICAL TO COGNITIVE

ERA 1 · 1776 - 1960S

MECHANICAL & KINETIC

DOMINANT FAILURE

Physical Burst

- **Nature:** Boilers bursting, gears shearing, belts severing.
- **Sensory:** The operator could smell smoke, hear screeching, and see steam leaks.
- **Understandability:** High. Governed by visible laws of mechanical stress.

ERA 2 · 1970S - 2020

CYBER-PHYSICAL (IT/OT)

DOMINANT FAILURE

Logic Hijacking

- **Nature:** PLC tampering, network worm propagation, Triton SIS overrides.
- **Sensory:** Invisible packets, spoofed SCADA telemetry, hidden weaponized code.
- **Understandability:** Moderate. Deterministic rules still governed logic execution.

ERA 3 · PRESENT & BEYOND

AUTONOMOUS & COGNITIVE

DOMINANT FAILURE

Hallucinatory Action

- **Nature:** Data poisoning, silent model drift, probabilistic write loops.
- **Sensory:** Non-existent. The system looks "optimal" right up until physical destruction.
- **Understandability:** Extremely low. Black box models lack mathematical proof cases for OT outcomes.

Why we cannot define modern risk: We are evaluating probabilistic, autonomous software with safety tools designed for steam boilers and relay ladders.

THE TEST FOR PLANT FLOOR AI

01 ARE YOU REDESIGNING THE WORK OR BOLTING IT ON?

If you're just feeding SCADA alerts into an LLM window, it is Phase 1 electricity. It will disappoint, and you will blame the model.

02 WHERE DOES OPERATIONAL CONTEXT ORIGINATE?

Can the system reference living asset topology and maintenance history? If not, the model is guessing with plant equipment.

03 WHAT IS THE RIGHT-HAND COLUMN OF YOUR RISK LEDGER?

Every breakthrough often creates a new vulnerability. Have you written down what happens IF the model hallucinates a setpoint change?

04 DOES THIS GIVE OPERATORS MORE TIME OR LESS?

If automated decisions compress response time is now below human reaction speed, you have traded resilience for risk acceptance.

03

PART THREE

THE CONTEXT THAT POWERS AI

AI does not fail because models are too small. It fails because it has zero operational context: no awareness of asset age, physical physics, or human workarounds.

WHAT MODELS DO NOT HAVE AND CANNOT EASILY INFER TODAY IN INDUSTRIAL ENVIROMENTS

Because; AI runs on operational context

01

Assets

What equipment exists, its configuration, its state, and how it behaves over time.

02

Relationships

How lines, systems and dependencies actually connect not how the diagram says.

03

History

What failed before, what fixed it, and the patterns sitting in between.

04

Process

How the work genuinely flows here, including the undocumented workarounds.

05

People

Who acts, who decides, and where human judgement sets the guardrail.

**None of these arrive
by making the model
bigger.**



Q&A?

Thank You!